



DAY 15: AUTO SCALING FOR HIGH AVAILABILITY

[General Interview Questions]

□ 1. What is Auto Scaling in AWS and why is it important?

Answer:

Auto Scaling automatically adjusts the number of EC2 instances based on traffic/load to ensure **high availability and cost-efficiency**.

□ 2. What are the types of Auto Scaling policies?

Answer:

- **Dynamic Scaling:** Responds to real-time metrics
- **Scheduled Scaling:** Based on time-based triggers
- **Predictive Scaling:** Uses machine learning to forecast future traffic and scale in advance

□ 3. What is the difference between a Launch Template and Launch Configuration?

Answer:

- **Launch Template:** Newer, supports versioning, more features
- **Launch Configuration:** Older, limited, being phased out

□ 4. What key parameters are required in a Launch Template?

Answer:

- AMI ID
- Instance Type
- Key Pair (optional)
- Security Group(s)
- IAM Role (optional)

□ 5. Can you update an Auto Scaling Group to use a new Launch Template version?

Answer:

Yes. You can **update the ASG** to reference a specific or latest Launch Template version at any time.

□ **6. What happens when minimum capacity = 0 in an ASG?**

Answer:

The Auto Scaling Group can **scale down to zero** EC2s during low traffic (useful for dev or batch workloads).

□ **7. How does Auto Scaling determine when to add or remove instances?**

Answer:

It uses **CloudWatch alarms** tied to metrics (CPU, memory, custom metrics) and scaling policies.

□ **8. How does Auto Scaling interact with Elastic Load Balancers?**

Answer:

Instances launched by ASG are **automatically registered** with the Load Balancer via **Target Groups**.

□ **9. What is a lifecycle hook in Auto Scaling?**

Answer:

It allows you to **pause instance launch or termination** to perform custom actions (e.g., install software, run scripts) before completing the process.

□ **10. Can Auto Scaling be used across multiple AZs?**

Answer:

Yes, ASG can be spread across **multiple Availability Zones** for high availability and zone-level fault tolerance.

□ **11. What happens if all subnets in an ASG become unavailable?**

Answer:

Auto Scaling will **not be able to launch instances** and will log errors unless other AZs/subnets are added.

□ **12. What is a Target Tracking Scaling policy?**

Answer:

A policy that **automatically adjusts capacity** to maintain a target metric (e.g., keep CPU at 50%).

□ 13. How does Predictive Scaling differ from Dynamic Scaling?

Answer:

Predictive Scaling **forecasts traffic** using machine learning and schedules scaling in advance; Dynamic reacts in real-time.

□ 14. Can Auto Scaling work with Spot Instances?

Answer:

Yes. You can configure an ASG to use **Spot Instances**, **On-Demand**, or a **mixed instances policy**.

□ 15. How would you implement Blue/Green deployment with Auto Scaling?

Answer:

Create **two ASGs** (blue and green), shift traffic gradually between them using **ALB target group weighting** or **Route 53 policies**.

◆ Q16. How is predictive scaling different from scheduled scaling?

- **Predictive Scaling:** Smart, automatic, based on **forecasted demand** using machine learning.
- **Scheduled Scaling:** Manual setup for known times